Blockhain Lab1 report

Setup:

- Docker install and running
- uv install and running
- strating container for regtest

```
tothandras@Toth-Air lab-01-bitcoin-regtest % docker compose up -d
[+] up 3/3
 ✔ Network lab-01-bitcoin-regtest_default     Created            0.0s
 ✔ Volume lab-01-bitcoin-regtest_bitcoin-data Created            0.0s
 ✔ Container lab01-bitcoind                   Started            0.1s
tothandras@Toth-Air lab-01-bitcoin-regtest % docker compose exec bitcoind bitcoin-cli getblockchain
info
{
  "chain": "regtest",
  "blocks": 0,
  "headers": 0,
  "bestblockhash": "0f9188f13cb7b2c71f2a335e3a4fc328bf5beb436012afca590b1a11466e2206",
  "difficulty": 4.656542373906925e-10,
  "time": 1296688602,
  "mediantime": 1296688602,
  "verificationprogress": 1,
  "initialblockdownload": true,
  "chainwork": "0000000000000000000000000000000000000000000000000000000000000002",
  "size_on_disk": 293,
  "pruned": false,
  "warnings": ""
}
```

- Create a wallet

```
tothandras@Toth-Air lab-01-bitcoin-regtest % bash ./scripts/setup.sh
== Waiting for bitcoind RPC ==
== Creating / loading wallet 'lab01' ==
   created new descriptor wallet
== Generating mining address ==
   bcrt1qmeaceggfrzqef2valeatr7qauajmvwl59lexxx
== Mining 101 blocks ==

== State ==
   block height        : 101
   spendable balance   : 50.00000000 BTC
   mature UTXOs        : 1
   UTXOs incl. immature: 1

Note the asymmetry: 101 coinbase outputs exist, exactly one is spendable.
Explain that in your report.

Next: docker compose exec bitcoind bitcoin-cli -rpcwallet=lab01 listunspent 100
```

Part A

"txid": "631ffddd5edfdc1e00fabd98e5dbf3f53074c043a6e5b65cec193fc3b40504a3",

"vout": 0,

"amount": 50.00000000,

"spendable": true,

"confirmations": 101,

"txouts": 101,

```
tothandras@Toth-Air lab-01-bitcoin-regtest % docker compose exec bitcoind bitcoin-cli -rpcwallet=la
b01 listunspent
[
  {
    "txid": "631ffddd5edfdc1e00fabd98e5dbf3f53074c043a6e5b65cec193fc3b40504a3",
    "vout": 0,
    "address": "bcrt1qpd5u5sr2at85968nlchag68rzkdqgh5ta484nl",
    "label": "",
    "scriptPubKey": "00140b69ca406aeacf42e8f3fe2fd468e3159a045e8b",
    "amount": 50.00000000,
    "confirmations": 101,
    "spendable": true,
    "solvable": true,
    "desc": "wpkh([c71508c2/84h/1h/0h/0/1]02487e35633283a3c7a8691f9adb34b375e1a2438021be4b3c7fab00a
f59dcb4bf)#8ppnvmk2",
    "parent_descs": [
      "wpkh(tpubD6NzVbkrYhZ4Xi3Aqwnz2f7ukqWZEqd7Je53VEQ683gLZGJTcQTwTJDovQvQRXPaVfRfrWrijfor2jEsWde
zpQ7eF5JEMwwZYpkRfpBqXim/84h/1h/0h/0/*)#y3qjw553"
    ],
    "safe": true
  }
]
```

Answer:

The second command counts every UTXO in the entire blockchain, in total 101 (101 * 50 = 5050), while the first one only looks at the outputs belonging to the wallet.

The node tracks all 101 block rewards because they exist on-chain, but the wallet hides the 100 immature ones because Bitcoin rules require 100 confirmations before a mines reward can be spent. Only block one is mature, so the wallet reports one spendable UTXO.

Part B – Building a raw transaction by hand

B1. Get the input

txid: 631ffddd5edfdc1e00fabd98e5dbf3f53074c043a6e5b65cec193fc3b40504a3

vout: 0

B2. Generate a destination and a change address

destination: bcrt1q7ysc7k455dkdtfauh3n7zufzd60g0pe7qnmlcr

change: bcrt1q9mfjgyn8puy43unqnd63cgumuchqu66qtc6g4k

B3. Compute the amounts

change = 50 - 10 - 0.0001 = 39.9999

B4. Create, sign and broadcast

signed hex:
02000000000101a30405b4c33f19ec5cb6e5a643c07430f5f3dbe598bdfa001edcdf5edd
fd1f630000000000fdffffff0200ca9a3b00000000160014f1218f5ab4a36cd5a7bcbc67e17
1226e9e87873ef0006bee000000001600142ed32412670f0958f2609b751c239be62e0e
6b400247304402203f7f5e9f218e0a8547601549bf2edde33c62dc2616035b0606d0879
972d6f0f602201555e7fa4b6b1d6bf574858d27913d99b8ebc495212a27662b42713c26
0ec476012102487e35633283a3c7a8691f9adb34b375e1a2438021be4b3c7fab00af59d
cb4bf00000000

After broadcast:

txid: 371897969187b3a6bc537b818bf5a2e6a7e2bfac013867ea7e1c67009aab488d

decodrawtransaction


```
{
  "txid": "371897969187b3a6bc537b818bf5a2e6a7e2bfac013867ea7e1c67009aab488d",
  "hash": "e04ef7049d6845abf6ba1d3a069e59df1eda71117e3e20af0b92ddb623fef4ec",
  "version": 2,
  "size": 222,
  "vsize": 141,
  "weight": 561,
  "locktime": 0,
  "vin": [
    {
      "txid": "631ffddd5edfdc1e00fabd98e5dbf3f53074c043a6e5b65cec193fc3b40504a3",
      "vout": 0,
      "scriptSig": {
        "asm": "",
        "hex": ""
      },
      "txinwitness": [
        "304402203f7f5e9f218e0a8547601549bf2edde33c62dc2616035b0606d0879972d6f0f602201555e7fa4b6b1d
6bf574858d27913d99b8ebc495212a27662b42713c260ec47601",
        "02487e35633283a3c7a8691f9adb34b375e1a2438021be4b3c7fab00af59dcb4bf"
      ],
      "sequence": 4294967293
    }
  ],
  "vout": [
    {
      "value": 10.00000000,
      "n": 0,
      "scriptPubKey": {
        "asm": "0 f1218f5ab4a36cd5a7bcbc67e171226e9e87873e",
        "desc": "addr(bcrt1q7ysc7k455dkdtfauh3n7zufzd60g0pe7qnmlcr)#epgwdyrt",
        "hex": "0014f1218f5ab4a36cd5a7bcbc67e171226e9e87873e",
        "address": "bcrt1q7ysc7k455dkdtfauh3n7zufzd60g0pe7qnmlcr",
        "type": "witness_v0_keyhash"
      }
    },
    {
      "value": 39.99990000,
      "n": 1,
      "scriptPubKey": {
        "asm": "0 2ed32412670f0958f2609b751c239be62e0e6b40",
        "desc": "addr(bcrt1q9mfjgyn8puy43unqnd63cgumuchqu66qtc6g4k)#yk2nn8lg",
        "hex": "00142ed32412670f0958f2609b751c239be62e0e6b40",
        "address": "bcrt1q9mfjgyn8puy43unqnd63cgumuchqu66qtc6g4k",
        "type": "witness_v0_keyhash"
      }
    }
  ]
}
```


Answer:

gettransaction: **-0.0010000**

change = 50 - 10 **- 0.0001** = 39.9999

The change is the same in both cases.

parser:

```
================================================================================
TRANSACTION
================================================================================
  txid            : 371897969187b3a6bc537b818bf5a2e6a7e2bfac013867ea7e1c67009aab488d
  wtxid           : e04ef7049d6845abf6ba1d3a069e59df1eda71117e3e20af0b92ddb623fef4ec
  version         : 2
  segwit          : True
  locktime        : 0  (no timelock)
  size            : 222 bytes
  stripped size   : 113 bytes  (witness excluded)
  weight          : 561 WU
  virtual size    : 141 vbytes  (fee rates are quoted per vbyte)

================================================================================
INPUTS (1)
================================================================================
  [0] byte offset 7
      prev txid     : 631ffddd5edfdc1e00fabd98e5dbf3f53074c043a6e5b65cec193fc3b40504a3
      prev vout     : 0
      scriptSig     : 0 bytes
      sequence      : 0xfffffffd  (RBF signalled, BIP-125)
      witness       : 2 stack item(s)
        [0] (71 B) 304402203f7f5e9f218e0a8547601549bf2edde33c62dc2616035b0606d0879972d6f0f602201555
e7fa4b6b1d6bf574858d27913d99b8ebc495212a27662b42713c260ec47601
        [1] (33 B) 02487e35633283a3c7a8691f9adb34b375e1a2438021be4b3c7fab00af59dcb4bf

================================================================================
OUTPUTS (2)
================================================================================
  [0] byte offset 49
      value         : 10.00000000 BTC  (1000000000 sat)
      type          : p2wpkh
      address       : bcrt1q7ysc7k455dkdtfauh3n7zufzd60g0pe7qnmlcr   [regtest]
      scriptPubKey  : 22 bytes
                      OP_0
                      <f1218f5ab4a36cd5a7bcbc67e171226e9e87873e>

  [1] byte offset 80
      value         : 39.99990000 BTC  (3999990000 sat)
      type          : p2wpkh
      address       : bcrt1q9mfjgyn8puy43unqnd63cgumuchqu66qtc6g4k   [regtest]
      scriptPubKey  : 22 bytes
                      OP_0
                      <2ed32412670f0958f2609b751c239be62e0e6b40>

================================================================================
TOTALS
================================================================================
  sum of outputs  : 4999990000 sat

  The FEE is not a field in this structure. It is
      sum(inputs) - sum(outputs)
  and the input values live in the outputs being spent, which are not
  part of this transaction. To compute the fee you must look up each
  previous output -- for example with:
      bitcoin-cli gettxout <prev_txid> <prev_vout>
```

Part C – Parsing the raw format

decoderrawtransaction:

"txid": "371897969187b3a6bc537b818bf5a2e6a7e2bfac013867ea7e1c67009aab488d",

txparser:

txid: 371897969187b3a6bc537b818bf5a2e6a7e2bfac013867ea7e1c67009aab488d

Part D – The deliberate mistake

txid: c6dac9fb0a554eabdd087de24e9dd937ae51e3d03ef616c6312c57158d9c39b7

vout: 0

destination: bcrt1qlmqfq94cl2tdlu0r3ja6qlydc9dsv822mqts62

change: bcrt1qavh5z38zf2cdczt3dt9ufwt82j5kauh2cqrfn9

50 – 10 + 40 (no change)

signed hex:
02000000000101b7399c8d15572c31c616f63ed0e351ae37d99d4ee27d08ddab4e550a
fbc9dac60000000000fdffffff0100ca9a3b00000000160014fec09016b8fa96dff1e38cbba
07c8dc15b061d4a024730440220121bb44456a5bbf76ba1477226bdf0a801c86ea93e4
06a7fbc4ae1747b17f43702203001d1c5774a5bfd0face42862ea62435966b3da17a18c1
02ab59b1e68d393d7012102070038ffe2cbd4fd2d17405ededece8ac0965277f8239ab5
3f6223bf5dbfff7c00000000

6786276bdf1d8f1d5dce635c12d15719a6bdae6510ef709bff102fe5d413d1d6
hash: 3b4296188005b4de229089c00c3271b01f2f2304d953007c12fca316d0680153

Question1:

Block value: 90 (50 + 40)


```
"vout": [
  {
    "value": 90.00000000,
    "n": 0,
    "scriptPubKey": {
      "asm": "0 d8b68cc378ae19f4ef952862dd4286cdcffde2dd",
      "desc": "addr(bcrt1qmzmgesmc4cvlfmu49p3d6s5xeh8lmcka83ekpm)#mfwj3v8u",
      "hex": "0014d8b68cc378ae19f4ef952862dd4286cdcffde2dd",
      "address": "bcrt1qmzmgesmc4cvlfmu49p3d6s5xeh8lmcka83ekpm",
      "type": "witness_v0_keyhash"
    }
```

Spent 50 BTC created one output of 10 BTC and fee of 40.



The 40 BTC is claimed by the miner who assembled the block. In Bitcoin there is no explicit fee field, it is rather calculated.

Fee = sum(input) – sum(output)

The transaction consumed the entire 50 BTC output and calculated a 40 BTC fee.

Question2:

Bitcoin consensus validation is strictly objective and does not infer user intent or enforce fee caps. A transaction is valid:

- referenced inputs exist
- signing of transaction by sender
- total output did not exceed total input

Question3:

Ethereum works like a regular bank account rather than cash bills (UTXO). Only the transferred amount is deducted. If we have 50 ETH and send 10, the state transition function simply subtracts 10 from the balance and credits 10 to the recipient. No concept of change. The remaining 40 ETH stays at the account. No routing back as change output. Transaction fee is handled differently and is determined by leftover funds and calculated via gas. Account based model does not use change outputs, so it is impossible to accidentally donate our balance to the miner.

Why does Bitcoin's fee mechanism have no explicit fee field, and what does this design buy?

Because the format omits an explicit fee field, the difference between inputs and outputs means the fee claimed by the miner in the block's coinbase transaction. This omission provides three major engineering and protocol advantages:

- No Conflicting Math: If an explicit "fee" field existed, nodes would constantly have to check if inputs = otuputs + fee.. Any mismatch would create conflicting data. Without a fee field, the rule is straightforward: outputs must not exceed inputs.
- Saves Block Space: Omitting an explicit 8-byte fee integer saves valuable data in every single transaction, keeping the blockchain smaller and faster to broadcast over the network.
- Simple Miner Payouts: Miners do not have to process extra scripts to collect their fees. They just sum the unspent remainder across all transactions in a block and credit the entire amount directly to themselves in the coinbase transaction.

You cannot calculate the fee from the raw transaction alone. Because input amounts are not stored inside the spending transaction, a node must check the UTXO database to look up how much money each referenced parent output was worth.